

E-23 3.2 requirement	Simple meaning	What should you validate?	Result / evidence to produce
Accurate and fit-for-use	Data should be correct and suitable for the model.	Check source-to-model reconciliation, errors, missing values, duplicates, invalid values and material biases.	Data Quality Test Results showing pass/fail, exceptions and impact.
Biases understood and managed	Data should not contain unexplained bias that could distort model results.	Compare distributions across relevant populations/segments; identify sampling or selection bias.	Data Bias Assessment with identified biases, impact and mitigation.
Relevant	Data should actually make sense for the model's purpose.	Check whether variables and historical data are relevant to the target/outcome being modelled.	Data Relevance Assessment with rationale for key data/variables.
Representative	Development data should represent the population where the model will be used.	Compare development sample vs. target/production population by key segments, products, geography, risk bands, etc.	Representativeness Analysis showing distributions and material differences.
Compliant	Data use must follow applicable rules and internal requirements.	Confirm data usage complies with applicable privacy, regulatory, data-ethics and internal requirements.	Compliance/Privacy confirmation or evidence of review by the appropriate control function.
Traceable	You should know exactly where the data came from.	Trace important fields from source systems through transformations to the final modelling dataset.	Data Lineage Validation / source-to-model traceability results.

E-23 3.2 requirement	Simple meaning	What should you validate?	Result / evidence to produce
Documented provenance	The origin and history of the data should be known.	Verify source, ownership, extraction, transformations and relevant history.	Data Provenance Assessment identifying sources and transformations.
Timely	Data should be sufficiently current for the model's purpose.	Check data dates, refresh frequency, stale records and whether the observation period remains appropriate.	Data Timeliness Test with aging/refresh results and exceptions.
Data quality checks	Data quality must be tested regularly.	Independently test missing values, outliers, duplicates, consistency, completeness and validity.	Detailed DQ Testing Workpaper with metrics, thresholds and pass/fail results.
Synthetic data	Artificially generated data needs additional controls.	If used, understand why it was used, how it was created and whether it changes model behaviour.	Synthetic Data Assessment including provenance, quality, representativeness and limitations.
Proxy data	Substitute data must be appropriate for what it represents.	Test whether the proxy reasonably represents the unavailable variable and whether it creates bias.	Proxy Data Assessment documenting rationale, correlation/relationship, limitations and impact.
Data cleansing	Changes made to raw data must be reasonable.	Review treatment of missing values, outliers, errors, duplicates, exclusions, imputation and transformations.	Data Cleansing Validation showing before/after results and assessment of treatments.
Unwanted bias / unfair outputs	Data bias can flow through into	Determine whether data characteristics could	Bias/Fairness Data Assessment , where

E-23 3.2 requirement	Simple meaning	What should you validate?	Result / evidence to produce
	model decisions.	create unintended or unfair model outcomes.	applicable to the model/use case.

A single **Data Validation Assessment** could contain the following evidence:

1. **Source-to-model reconciliation**
2. **Completeness and missing-value analysis**
3. **Outlier and invalid-value testing**
4. **Duplicate and consistency testing**
5. **Development vs. target population representativeness analysis**
6. **Data lineage/provenance verification**
7. **Data timeliness assessment**
8. **Data cleansing/transformation review**
9. **Synthetic/proxy data assessment**, if applicable
10. **Data bias assessment**, where applicable
11. **Exceptions/findings and their model impact**
12. **Overall conclusion: Pass / Pass with Issues / Fail**, or your institution's equivalent.